The Region of Convergence

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Key Concepts

1) The region of convergence (ROC) is the set of values z for which the infinite sum $X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$ converges.

2) $X(z)$ is a power series in z. Convergence is assured for values of z where the series is absolutely summable.

    a) Convergence depends only on $|z|$, not the phase.

    b) The ROC involves rings in the z-plane.

3) Two signals can have identical z-transforms, but are differentiated by having different ROCs.

4) The z-transform does not exist if there are no values of z for which the sum converges.

Video Screenshots

Region of Convergence (ROC)

ROC: set of z for which the z transform of a signal $x[n]$ converges (exists)

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

Case 1: Delay $w[n] = \delta[n-n_0]$

$$W(z) = \sum_{n=-\infty}^{\infty} \delta[n-n_0] z^{-n} = z^{-n_0}$$

excludes $z=0$ for $n_0>0$
$z=\infty$ for $n_0<0$

$$\delta[n-n_0] \overset{z}{\longleftrightarrow} z^{-n_0} \quad \forall \begin{matrix} z\neq 0 \text{ if } n_0>0 \\ z\neq\infty \text{ if } n_0<0 \end{matrix}$$

Case 2: $x[n] = \alpha^n u[n]$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]\, z^{-n}$$

$$= \sum_{n=0}^{\infty} \alpha^n z^{-n} = \sum_{n=0}^{\infty} (\alpha z^{-1})^n$$

$$= \frac{1}{1 - \alpha z^{-1}} \qquad \text{provided } |\alpha z^{-1}| < 1$$

$$|z| > |\alpha|$$

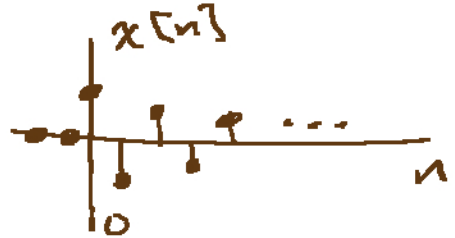
if $|\alpha z^{-1}| \geq 1$, $X(z)$ does not converge



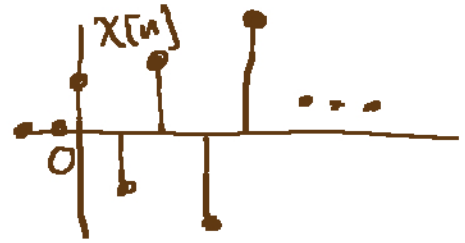
$0 < \alpha < 1$



$\alpha > 1$



$-1 < \alpha < 0$



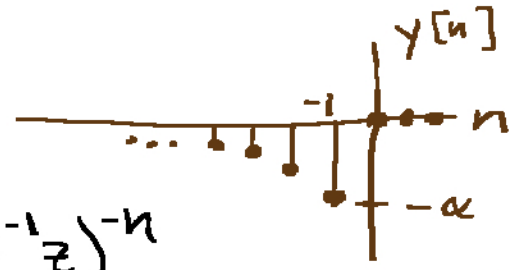
$\alpha < -1$

$$\alpha^n u[n] \overset{z}{\longleftrightarrow} \frac{1}{1 - \alpha z^{-1}} = \frac{z}{z - \alpha} \qquad |z| > |\alpha|$$



ROC: $|z| > |\alpha|$

Case 3: $y[n] = -\alpha^n u[-n-1]$



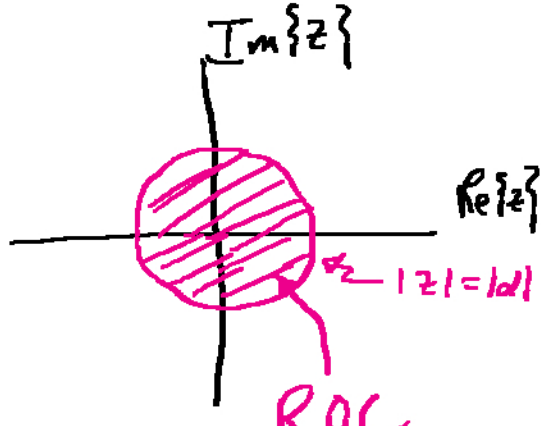
$0 < \alpha < 1$

$$Y(z) = \sum_{n=-\infty}^{\infty} y[n] z^{-n} = \sum_{n=-\infty}^{-1} -\alpha^n z^{-n} = -\sum_{n=-\infty}^{-1} (\alpha^{-1} z)^{-n}$$

$$= -\sum_{l=1}^{\infty} (\alpha^{-1} z)^{l} = -\sum_{l=0}^{\infty} (\alpha^{-1} z)^{l} + 1 = \frac{-1}{1-\alpha^{-1} z} + 1 \quad \text{for } |\alpha^{-1} z| < 1$$

$$= \frac{\alpha}{z-\alpha} + 1 = \frac{z}{z-\alpha} \quad \left(\text{or } \frac{1}{1-\alpha z^{-1}}\right) \qquad |z| < |\alpha|$$

$$-\alpha^n u[-n-1] \overset{z}{\longleftrightarrow} \frac{1}{1-\alpha z^{-1}} = \frac{z}{z-\alpha} \qquad |z| < |\alpha|$$



X(z) (Case 2) is identical to Y(z)!

Differ in the ROC - z transforms nonunique w/o ROC

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]\, z^{-n}$$ power series in z

Convergence $\Rightarrow$ absolutely summable

Require $$\sum_{n=-\infty}^{\infty} |x[n]\, z^{-n}| = \sum_{n=-\infty}^{\infty} |x[n]|\, |z|^{-n} < \infty$$

ROC depends on $|z|$ - circles/rings in z-plane

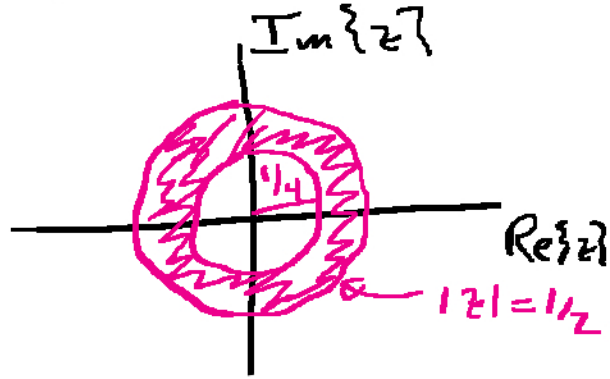
Example: $g[n] = (1/4)^n u[n] - (1/2)^n u[-n-1]$

$(1/4)^n u[n] \overset{z}{\longleftrightarrow} \frac{z}{z-1/4}, \quad |z| > 1/4$

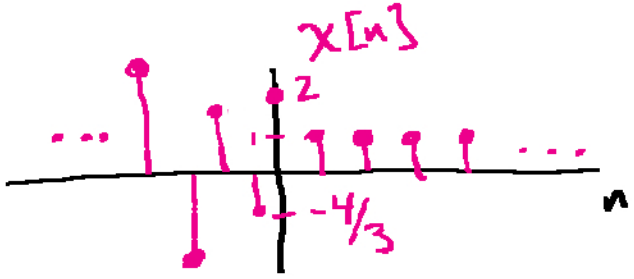
$-(1/2)^n u[-n-1] \overset{z}{\longleftrightarrow} \frac{z}{z-1/2}, \quad |z| < 1/2$

$$G(z) = \frac{z}{z-1/4} + \frac{z}{z-1/2}, \quad 1/4 < |z| < 1/2$$

$$= \frac{2z^2 - \frac{3}{4}z}{(z-1/4)(z-1/2)},$$



Example: $x[n] = u[n] + (-3/4)^n u[-n]$



To use transform pairs, rewrite

$$x[n] = u[n] + \delta[n] - -(-3/4)^n u[-n-1]$$

$$u[n] \overset{z}{\longleftrightarrow} \frac{z}{z-1}, \quad |z|>1$$

$$\delta[n] \overset{z}{\longleftrightarrow} 1$$

$$-(-3/4)^n u[-n-1] \overset{z}{\longleftrightarrow} \frac{z}{z+3/4}, \quad |z|<3/4$$

incompatible

$X(z)$ does not exist!

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